Predicting Offensive Play-Calling and Success in the NFL Using Machine Learning Techniques

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***Abstract*—Machine learning has become an increasingly valuable tool for analyzing large quantities of data and gaining meaningful insights into professional sports. This study applies Logistic Regression and Random Forest classification models to National Football League (NFL) data from the 2009 through 2016 seasons to investigate questions related to offensive play-calling and success. First, the study examines whether an offense will call a run or a pass play based on basic situational factors of the game. Second, it identifies which features affect an offense's success the most, where success is defined as a play with a positive Expected Points Added (EPA). Lastly, it determines whether NFL teams consistently select the play with the highest probability of success. The models were trained using variables that describe the game situation, including down, field position, yards to go, time remaining, score differential, etc. The models' performance was evaluated using confusion matrices, classification accuracy, ROC curves, AUC values, and variable importance analysis. The results suggest that offensive play selection can be predicted with moderate success, while predicting offensive success requires additional factors. Overall, this study demonstrates how machine learning techniques can provide a deeper understanding of offensive strategy and data-driven decision-making in professional football.**

Keywords—Machine Learning, Logistic Regression, Random Forest, National Football League (NFL), Sports Analytics, Expected Points Added (EPA), Offensive Play Calling, Predictive Modeling

I. Introduction

Statistical modeling and data analysis have been increasingly utilized in sports, whether that be at the professional level or in high school. Across all sports, there is a large amount of data being generated every game. That data is then analyzed by the organization to improve the team and help coaches make better strategic decisions about their gameplay. Advances in machine learning and statistical modeling have made it possible to find hidden relationships and patterns within these large, and most of the time extremely messy, datasets. To be more specific, in the National Football League (NFL), these patterns can give the coaching staff guidance for offensive play calling, defensive formations, or any other factors that could lead to successful outcomes. One of the simplest and most important decisions an offensive coordinator makes before every play is whether to pass or run the ball. When making this decision, coaches consider several factors, such as current down, field position, time left, score, etc. Although there are other factors that can not be effectively measured, there is evidence that there are consistent patterns within this data that link certain factors to successful plays.

In addition to understanding play selection, it is also important to determine what variables contribute to offensive success. Before that, we also need to determine what success means in terms of a singular play. There is a statistic called Expected Points Added, or EPA. It is a measurement of how much a single play increases or decreases the likelihood of scoring. This metric is widely used to measure the impact of an individual play, making it a suitable candidate for use in our study.

This study focuses on three main research questions using NFL Play-by-Play data spanning from the 2009 season to the 2016 season. First, can offensive play type be predicted based on situational game variables? Second, what specific factors contribute the most to a successful play? Third, do coaches consistently choose the play with the highest expected success? To answer these questions, Logistic Regression and Random Forest models are created and then evaluated. Logistic regression models are created first to give a baseline answer to the question. Then, the random forest models are developed to analyze the more complex and nonlinear relationships hidden within the dataset. The random forest models also allow us to dive into feature importance to determine what variables affect the model the most. The performance of each model was tested using accuracy, confusion matrices, ROC

(Receiver Operating Characteristic) curves, and AUC (Area Under the Curve) values.

After completing the analysis of these models, we now have some insight into the predictability of offensive plays. We can also identify what variables contribute the most to successful plays and evaluate how often offensive coordinators choose the play with the highest expected probability of success. This shows us how machine learning and statistical models can be applied to professional sports to better understand data-driven decision-making and performance in football.

II. Background

The data used in this study was retrieved from Kaggle, an online platform for machine learning practitioners and data scientists to find credible data to perform analysis on. The dataset is called Detailed NFL Play-by-Play Data, and it was published by Max Horowitz [1]. There are actually three separate datasets available containing different timeframes: 2009-2016, 2009-2017, and 2009-2018. For this project, we will be using the 2009-2016 dataset because it contains an adequate amount of observations to train and test the models. Each observation in this dataset represents a singular play and includes many different variables such as Qtr, Time, Down, PlayType, and various probability statistics (including EPA). That is what makes this dataset so perfect for this type of analysis.

However, before the analysis can begin, we need to preprocess or clean the dataset to ensure it is formatted correctly. First, the PlayType variable was filtered to only contain the values “Pass” and “Run”. This ensures that we are only looking at plays where a pass or a run play was attempted. For this analysis, we did not need to consider any special teams plays such as kickoffs or punts. Second, any overtime plays were filtered out of the dataset. This is because the decision-making process in an overtime situation is much different than regular gameplay. Third, the PlayType variable is changed to a binary variable (zeros and ones) to make calculations easier. Fourth, the columns were renamed to provide more clarity in the report. Fifth, all missing values were removed from the dataset because incomplete observations do not help our model’s performance.

One of the variables utilized in this analysis was Expected Points Added, or EPA, which is a widely used advanced metric that can be seen as “yards with context” [2]. This variable helps a team define what a successful play is by asking, “Did we get closer or further from scoring?” [2]. A positive value indicates that the play improved the offense’s chances of scoring; meanwhile, a negative value indicates that the play decreased the offense’s chances of scoring. Because EPA considers down, distance, position, and other pieces of the game, it provides a more comprehensive measure of offensive performance.

In this study, EPA is utilized in response to the second question. It provides a way for us to define what a successful play is by defining it as an EPA value greater than zero. Using this variable as a binary classification allows us to predict whether a play would produce a positive outcome.

Lastly, there are two supervised machine learning techniques used in this study: Logistic Regression and Random Forest. Logistic Regression is a classification technique that is designed for binary outcome variables. That is why the PlayType variable was changed to be a binary variable so that it would fit as the target variable in a Logistic Regression model. Instead of predicting a continuous variable, it estimates the probability of an observation fitting into one of two possible outcomes [3]. For this report, this model gives us a baseline model to see the simple structure of the relationship between the target variable and the predictors. Meanwhile, Random Forest is an ensemble-based learning method that uses bootstrapping to create an unrelated forest of decision trees to produce a classification [4]. In addition to classification, Random Forest models also provide feature importance values that can help identify which game factors strongly influence offensive play selection and success.

The performance of the models was evaluated using multiple different metrics. However, before that could happen, the data needed to be split into training and testing datasets. Using a command in Scikit-Learn, 70% of the data was placed into training, and the other 30% was placed into testing. Then, the machine learning models were compared using accuracy, confusion matrices, ROC (Receiver Operating Characteristic) curves, and AUC (Area Under the Curve) values. ROC curves are a useful way to assess the accuracy of model predictions by plotting sensitivity vs. specificity [5]. AUC represents the probability that the prediction is correct [5]. Lastly, in the Random Forest models, hyperparameters were selected using the GridSearchCV function with five-fold cross-validation specified to identify what combinations of estimators and features created the model with the best accuracy.

III. Methods

The project was performed using supervised machine learning techniques on play-by-play data sourced from the NFL. Furthermore, analysis was performed using Python with the pandas, NumPy, seaborn, matplotlib, and scikit-learn libraries. Two classification algorithms were utilized: Logistic Regression and Random Forest. The main objective of this project was to answer these three research questions:

1. Can we predict what play type the offense will call? Run or pass?
2. What factors contribute the most to successful offensive plays?
3. Do teams consistently choose the play with the highest expected success?

A. Data Preprocessing and Cleaning

The original NFL Play-by-Play dataset contained over 350,000 observations with 102 columns, and some of the data is missing or untidy. Therefore, the data required a few preprocessing steps before beginning the analysis. The dataset was first filtered to only include offensive rushing and passing plays. Certain plays such as special teams, timeouts, and penalties were removed as they were not pertinent to the study. Also, overtime observations were removed by limiting the dataset to regulation play by only allowing quarters one through four. This was done because the decision-making process is different in overtime play, as the coaching staff is more likely to be stressed and looking for any way to score points. Most columns were removed to simplify the model creation process. The columns that were retained were renamed for simplification purposes. Lastly, any missing values were removed from the dataset as missing data can affect the training process in the models.

For the first research question, the target variable was PlayType, which was classified as a string variable containing either "Pass" or "Run". As previously discussed, a logistic regression model works with binary target variables; therefore, the PlayType variable was changed to a binary variable. Runs were now classified as zeros and passes were classified as ones. Predictor variables included Down, YdsToGo, YrdLine, GoalToGo, Qtr, Time, ScoreDiff, OffTimeouts_Pre, and Season. Definitions for these variables can be found in the Data Dictionary.

For the second and third research questions, the target variable, named Success, was created using EPA. This variable was a binary variable that determined whether a play was successful or not. Plays with an EPA greater than zero were considered to be successful, while plays with an EPA of zero or less were considered to be unsuccessful. Predictor variables were PlayType, Down, YdsToGo, YrdLine, GoalToGo, Qtr, Time, ScoreDiff, OffTimeouts_Pre, and Season. Definitions for these variables can be found in the Data Dictionary.

B. Question 1: Predicting Offensive Play Types

The first objective was to determine whether offensive plays could be predicted using basic game situational factors. The cleaned dataset was divided into training and testing datasets using a 70/30 split, with a random state specified to ensure the same outputs could be reproduced during testing. The split was done using the train_test_split function from the scikit-learn package.

Then, a Logistic Regression model was created using the selected situational variables described before to estimate the probability that a play would be a pass. This was done by using the LogisticRegression command from the scikit-learn library [6]. After completing the model, its performance was evaluated using a confusion matrix, accuracy score, and a receiver operating characteristic (ROC) curve with a corresponding AUC score. All of these evaluations were sourced from the metrics library in scikit-learn [7].

Finally, a Random Forest Classification model was performed using those same predictors to try and uncover any nonlinear relationships, as well as see if the performance of the model improves from our base logistic regression model. This model was created using the RandomForestClassifier function from the scikit-learn package [8]. GridSearchCV, another function located in the scikit-learn library, with five-fold cross-validation, was utilized to evaluate multiple combinations of estimators and features. Once the model finds the combination with the highest accuracy, it is evaluated on the testing dataset to determine the accuracy.

C. Question 2: Predicting Offensive Success

The second objective was to identify what situational factors contribute the most to successful offensive plays. The cleaned dataset was, again, divided into training and testing datasets using the same 70/30 split, and the same random state was specified for reproducibility.

Another Logistic Regression model was used to estimate the probability that a play would be a successful play, based on the earlier definition of success based on Expected Points Added (EPA). Following the same process as the previous logistic regression model, it was then evaluated using a confusion matrix, accuracy score, ROC curve, and AUC score.

Lastly, another Random Forest Classification model was performed to identify nonlinear relationships and attempt to improve upon the baseline logistic regression model. However, for this model, the evaluation process was modified. An accuracy score was still calculated to compare to the logistic regression model [9]. After that, an importance plot was created using the feature_importances_ attribute of the RandomForestClassifier function to display the importance of each variable to the model in a seaborn barplot [8]. This is where we identify which aspects of the game are most influential to offensive success.

D. Question 3: Evaluating Offensive Decision-Making

The third and final objective was to determine whether a coach's gut decision usually follows what the statistics say is the play with the highest expected probability of success. The Random Forest model developed for the second question was used for this analysis because it provided probability estimates for offensive success.

To begin, two copies of the testing dataset used in the second question were created. In the first copy, named runs, the PlayType variable was changed to a zero to indicate that every observation was a rushing play. In the second copy, named passes, the PlayType variable was changed to a one to indicate that every observation was a passing play. All other

variables in these datasets were left unchanged. This allows us to compute what the probability of success would be for every observation if it was a pass and a run. The Random Forest model was then used to estimate these probabilities for both hypothetical scenarios.

For each observation, the predicted success probabilities for rushing and passing were compared in a pandas DataFrame. Once compared, the play with the highest probability was then assigned as the "recommended play". This can then be compared to the actual play run by the offense. Finally, another column was created called "Match", which is a boolean variable that determines whether the recommended play matched the actual play. The percentage of observations where a match occurred was calculated to estimate how frequently offensive coordinators selected the play that the model predicted would have the greatest chance of success.

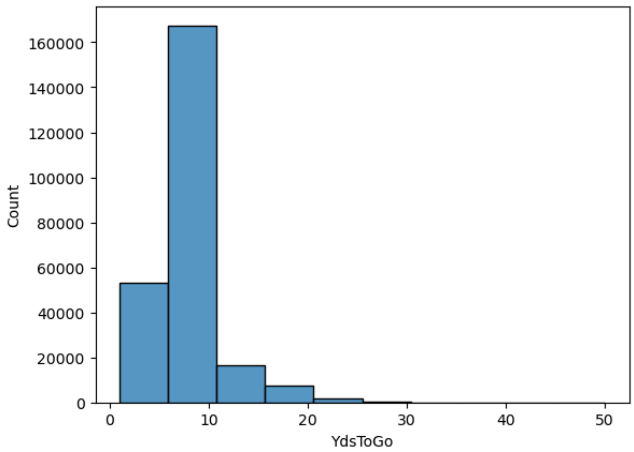
IV. Results

This section presents the results of the exploratory data analysis (EDA) and machine learning models produced for our three research questions. First, descriptive statistics, correlational analysis, and visualizations were created to give a better understanding of how these variables correlate to each other. Then, Logistic Regression and Random Forest models were developed and evaluated. Lastly, the results of the decision-making analysis are discussed.

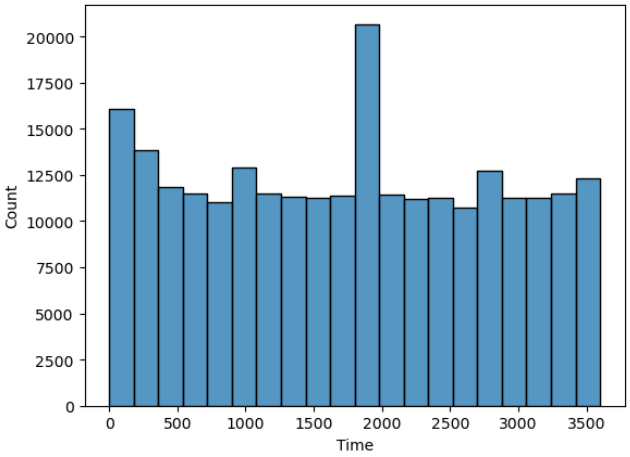
A. Exploratory Data Analysis (EDA)

Descriptive statistics were first computed to summarize the variables included in the analysis. Variables such as Down, YdsToGo, YrdLine, and ScoreDiff show that this dataset contains a vast amount of plays that cover a variety of different game situations. This proves that this dataset is a good fit for our research analysis.

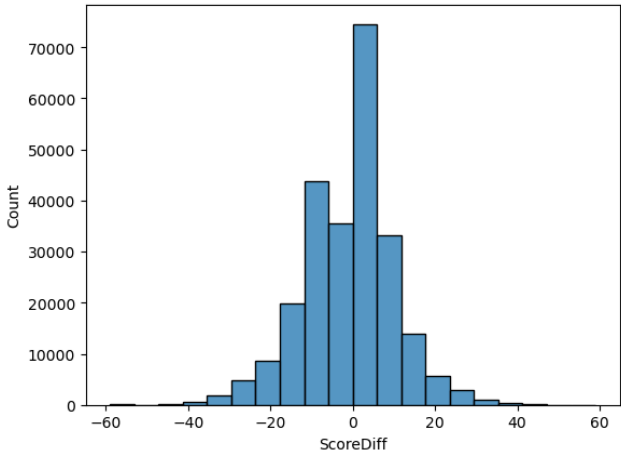
The first visualization is a histogram of the YdsToGo variable. This variable represents the number of yards needed to earn a first down. Therefore, it is expected for this histogram's peak to be focused around 10 yards because when a team receives possession of the ball, the line of scrimmage is placed 10 yards behind the first down marker. However, the range of this histogram runs from 0 yards to 30 yards. This suggests that many offensive drives consist of first-and-ten plays; meanwhile, there were other plays present that showed more deviation from the normal start of a set of downs, such as short third-down conversions or long-shot second downs.



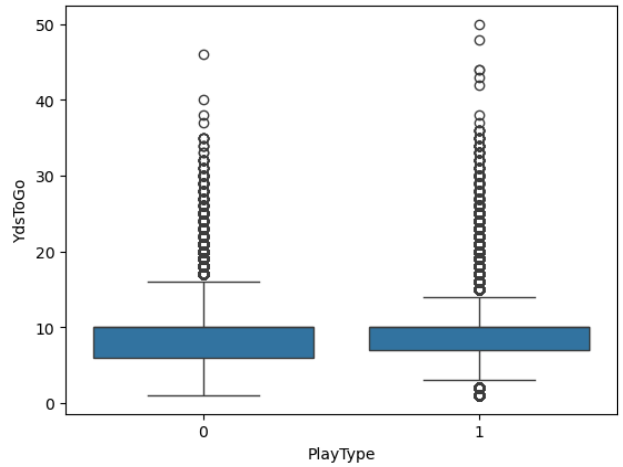
The second visualization shows the distribution of the Time variable. A variable that represents the number of seconds left in the game. The expectation of this histogram would be an even distribution across all 20 bins. This is mostly true, except there is a higher number of observations around 2000 seconds. This means that there is a greater number of observations that occur right before half-time in this dataset. However, the rest of the dataset is spread evenly across all other parts of the game.



Moving on to the third visualization, another histogram focusing on the ScoreDiff variable. This variable describes the difference in scores between the offensive and defensive teams at the start of the play. For specificity, this variable is calculated by taking the offensive team's score and subtracting the defensive team's score from it. This means that a positive value indicates the offensive team is winning, while a negative value indicates the defensive team is winning. Most plays occurred while games were relatively close; however, there are larger score margins represented in this dataset.



The final chart created in this analysis compares the distribution of YdsToGo for passing and rushing plays using boxplots. When looking at the rushing plays (zeros), the range is wider, which suggests that rushing plays are more called in for longer yardage situations. On the other hand, passing plays are used in shorter range situations. However, there are many outliers in both boxplots, suggesting that both play types are utilized in both situations.



Finally, a correlation matrix was developed to examine the specific relationships among each of the predictors. Most correlations are relatively uniform, especially the variables that contain categorical variables, such as Qtr, Down, and Season. As for the actual correlation values, none of the relationships have a value close to 1 or -1, which suggests that multicollinearity was not present within the dataset.

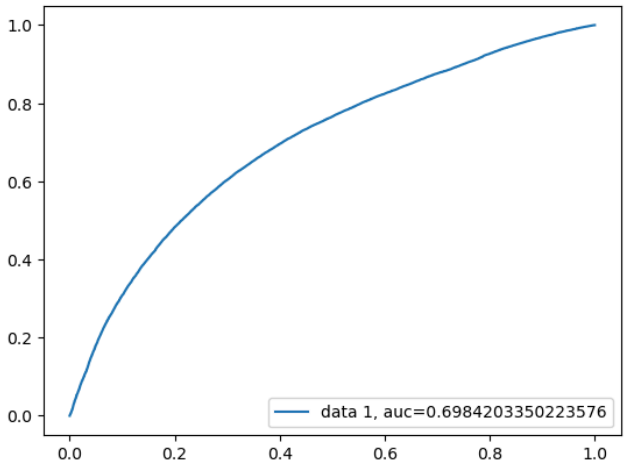
B. Question 1: Predicting Offensive Play Types

To begin, a Logistic Regression model was trained to classify offensive plays as either runs or passes based on game situational factors. The model's performance was then evaluated using multiple different metrics, such as a confusion matrix, accuracy score, ROC curve, and an AUC score.

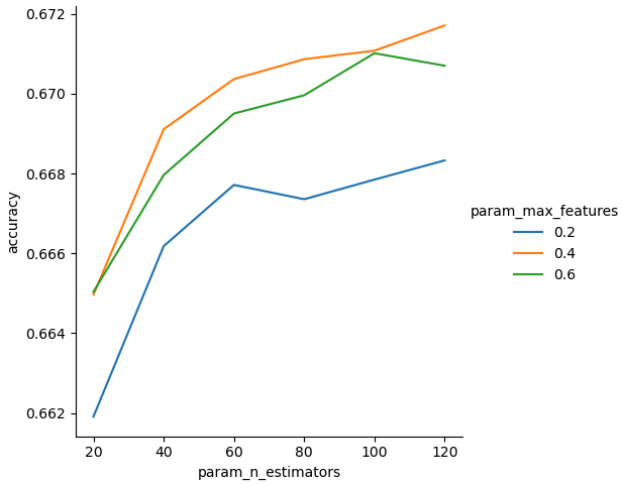
The confusion matrix displays the number of correctly and incorrectly classified plays. According to this matrix, the model was able to classify a large number of plays correctly. However, there is still some overlap between the two classes. This reflects the uncertainty of offensive play calling. The model achieved an overall classification accuracy of about 65%, suggesting that these situational variables do provide some reasonable ability to predict offensive play types.

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array([[17553, 14384],
       [11223, 30929]])
```

The ROC curve further demonstrates the model's ability to distinguish between rushing and passing plays because the curve does not hug the main diagonal of the graph. This tells us that the model does a substantially better job of predicting play types than random guessing would. Furthermore, the resulting AUC score of about 0.7 suggests that the model's ability to predict play type is acceptable.



Next, a Random Forest Classifier model was trained using GridSearchCV with five-fold cross-validation to identify what the best combination of estimators and features creates the best classification model. The line chart pictured below shows the accuracy for each of the combinations attempted by the model. The highest accuracy was achieved by using 120 estimators and 0.4 max features, which generated an accuracy score of about 67%.



When evaluating on the validation dataset, the Random Forest model performed better than the Logistic Regression model. This suggests that nonlinear relationships between play type and situational game factors improve the predictive performance of the model.

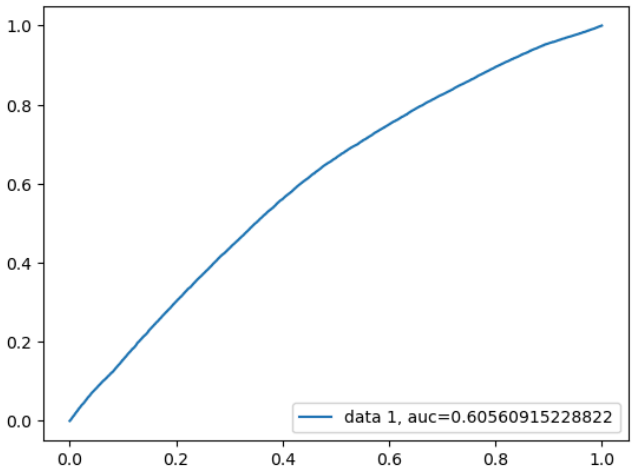
C. Question 2: Predicting Offensive Success

Following the same procedure as the first question, a Logistic Regression model was trained. However, this model was trained to predict whether a play would be successful or not. The definition of success is defined by the EPA, or Expected Points Added, of the play. A play with a positive EPA is deemed successful, while a play with a negative EPA is deemed unsuccessful.

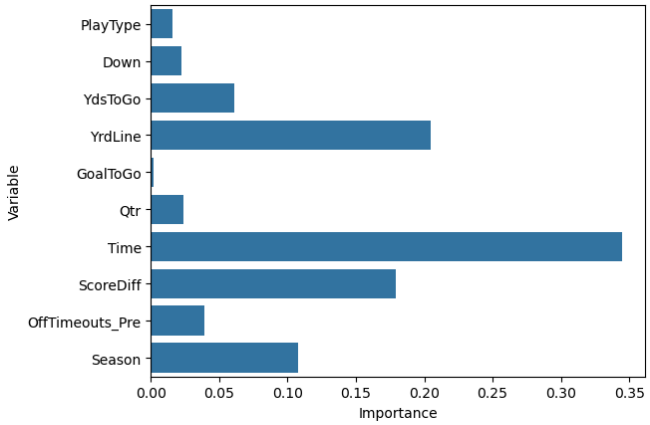
The Logistic Regression model also produced a confusion matrix that summarizes how the model classified each observation in the dataset. Compared to the previous research question, predicting offensive success proved to be more difficult than predicting the play type. This model achieved an overall classification accuracy of about 59%. This suggests that the model has a moderate predictive ability to predict offensive success.

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array([[34555,  7750],
       [22938,  8813]])
```

The ROC curve also corroborates the idea that this model was not able to predict success as well as the previous Logistic Regression model. The ROC curve in this graph is much closer to the main diagonal, which puts this model's predictive power closer to random guessing. The AUC score for this model is about 0.61. The modest classification accuracy of this model suggests that offensive success depends on factors that are not included in our list of predictors.



Then, another Random Forest model was trained to predict whether or not a play will be successful. GridSearchCV was used again with a five-fold cross-validation to identify the best combination of factors. Similar to the previous Random Forest model, the most successful combination was 120 estimators and 0.4 max features. This particular combination achieved a classification accuracy of about 58%. However, we are more interested in what specific variables affect the model the most. Therefore, an importance plot was created to show how important each variable is to the predictive power of the Random Forest model. According to the plot, the most important variables are Time, YrdLine, and ScoreDiff. This tells us the score, field position, and time left in the game are the factors that affect a team's chances of success the most. Other variables, such as GoalToGo, PlayType, and Down, do not affect a team's chances as much.



This time, the Logistic Regression model achieved a higher classification accuracy than the Random Forest model, but by only about 1%. This suggests that there are other factors that contribute to the success of a play that are not present in our list of predictors. However, it is still worth noting that both models did have a moderate ability to predict success. It is possible that the models need more variables to increase their predictive power.

D. Question 3: Evaluating Offensive Decision-Making

The final research question asks us to compare actual offensive play calls with the recommended play generated by the Random Forest model produced to answer the second question. For each observation in the validation dataset, the probability of success was calculated for a rushing play and a passing play. The play with the higher probability is recorded as the recommended play, as it is the play type with the highest probability of success.

A comparison table was generated containing the actual play, the recommended play, the run success probability, the pass success probability, and a match variable. This match variable is a boolean variable that shows whether the actual play and recommended play match.

	Actual Play	Recommended Play	Run Success Probability	Pass Success Probability	Match
0	1	1	0.391667	0.600000	True
1	1	0	0.333333	0.158333	False
2	0	1	0.125000	0.233333	False
3	1	0	0.208333	0.125000	False
4	0	1	0.333333	0.391667	False
...	...	...	...	...	...
74051	0	1	0.491667	0.525000	False
74052	1	1	0.566667	0.600000	True
74053	1	1	0.400000	0.683333	True
74054	1	1	0.216667	0.400000	True
74055	1	1	0.280556	0.500000	True

Lastly, the percentage of observations in which the model correctly matched the actual offensive call to the recommended play generated by the success model. This value provides a general estimate of how frequently NFL coaches select the play that the model predicted would have the highest probability of success. This percentage was calculated and found to be about 48%. Differences between the model recommendations and actual play calls indicate that coaches are using other data not included in the dataset to make play-calling decisions. This data could represent a wide range of information, including defensive formations, personal player profiles, or game-specific strategies.

V. Conclusion

This study investigated three separate research questions related to offensive play-calling decisions in the National Football League (NFL) using machine learning techniques and play-by-play data sourced from the 2009 through 2016 seasons. Logistic Regression and Random Forest classification models were produced to predict offensive play type, identify important factors that contribute to successful offensive plays, and determine whether NFL teams consistently chose the play with the highest predicted probability of success.

The results indicated that offensive play type can be predicted using certain situational variables such as down, yards to go, field position, time remaining, and score differential. The Random Forest model outperformed the Logistic Regression model, suggesting that nonlinear relationships between the predictors improve the predictive power of the model.

On the other hand, predicting offensive success proved to be more difficult. Both the Logistic Regression and Random Forest models performed better than random guessing, but both achieved about the same classification accuracy. The decreased classification accuracy suggests that the outcome of an individual play depends on many factors, including ones that are not included in the dataset. Variable importance analysis was conducted, and it suggested that time remaining, field position, and score margin contributed the most to an individual play's success. Although no single variable can completely explain the outcome of the play.

The final research question guided us to compare actual offensive plays with the recommendations given by the Random Forest model trained for play success. The results indicated NFL teams select the recommended offensive play about half of the time. These differences reflect that there is other information unavailable to the model that would help improve the model's predictive power.

The models developed for this project provided meaningful insights; however, several limitations should be considered. First, the training and validation dataset relied exclusively on a small set of features. It is possible that there are other contextual factors that would influence coaching decisions during a game. Additionally, using EPA to define what a successful play is does simplify a rather complicated concept into a binary value.

Future work could expand this study by incorporating more information in the training and testing datasets. By adding more information, such as defensive strategies, weather conditions, or more advanced football metrics, the machine learning techniques could become more refined and increase their classification accuracy. Furthermore, more sophisticated models, such as XGBoost, could also be used to evaluate the questions to see if performance can be improved. Despite the limitations, this research demonstrates that offensive decision-making can be evaluated and quantified using machine learning techniques, further highlighting the growth of data-driven decisions in modern sports.

Data Dictionary

Column Name	Description
Down	Down of the given play
YdsToGo	Yards to go for a first down
YrdLine	The distance to the opponents endzone, between 1 and 99
GoalToGo	1 if the play is in a goal down situation else 0 (when the offensive team is close to the opponents endzone, and-goal situations)
Qtr	Quarter of the game
Time	Time remaining in game in seconds
ScoreDiff	The difference in score (offensive score - defensive score)
OffTimeouts_Pre	The number of timeouts remaining for the offensive team at the start of the play
Season	The season that the game was played in
PlayType	The type of play (pass or run)
EPA	The expected points added with respect to the offensive team considering the result of the play
Success	1 if the play is deemed successful (EPA > 0) else 0

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